

Summary

Electronics Engineering student specialising in VLSI, with hands-on experience in UAV systems, embedded hardware, and PCB design. Adept at integrating LiFi (Light Fidelity) with optical fiber backhauls to create secure, zero-EMI, high-bandwidth communication networks. Interested in cryogenic electronics and its applications in quantum computing hardware.

Employment History

- 2026
- 📖

Design and Analysis of Laser Based LiFi Network for Indoor Wireless Communication - Prof. Ravi Maddila, MNIT Jaipur
 - Designed a laser-based Light Fidelity (LiFi) communication system to improve indoor wireless communication speed, increase signal quality, and reduce the Bit Error Rate (BER).
 - Analyzed system performance using OptiSystem software by simulating components such as a Continuous Wave (CW) Laser, Mach Zehnder Modulator (MZM), and a Low Pass Chebyshev Filter.

Experience & Projects

- 2025
- 📖

Lead Developer - F450 Multicopter Drone
 - Built a complete F450 quadcopter from scratch, integrating hardware components and tuning PID algorithms for stable flight performance and signal reliability. Achieved 20-minute flight time.
 - Integrated computer vision (OpenCV, Python) with autonomous flight logic via ArduPilot, enabling real-time face recognition and subject tracking (90% accuracy) for disaster-response use cases.

Hackathons & Achievements

- 2026
- 📖

State Finalist - Technovation Hackathon 2026

Led a team to design and build a face-detection drone using carbon fibre and 3D printed components, competing at state level.
- 2025
- 📖

Internal Smart India Hackathon (SIH) - Developer

Deployed a lightweight web architecture for a rural EdTech platform to improve education access in low-connectivity areas.

Skills, Tools & Certifications

- Technical
- 📖

Raspberry Pi, ESP32, Arduino, UAV Architecture, PCB Design, Computer Vision, Embedded Systems Programming, Python, C++
- Tools
- 📖

Altium Designer, EasyEDA, SolidWorks, Tinkercad, OptiSystem, Zemax
- Certifications
- 📖

Altium Education PCB Design

Leadership & Interests

- Event Role
- 📖

PR & Event Representative - Eventique, RRU
- Interests
- 📖

Philosophy, Art History, Precision Embroidery, Abstract Sketching